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5. An experiment was carried out by students to investigate the mean speed of a stack of 6 cake cases falling from different heights.
The weight of 6 identical cake cases was measured 5 times.
Their results are shown below in **Table 1** and there were no anomalies.

Table 1

Measurement	1	2	3	4	5
Weight of 6 cake cases (N)	0.036	0.032	0.033	0.034	0.030

(a) (i) Calculate the mean weight of the 6 cake cases. [1]

Mean weight = ... N

(ii) Use the equation:

$$\text{uncertainty} = \frac{\text{maximum value} - \text{minimum value}}{2}$$

to calculate the uncertainty in the mean weight of the 6 cake cases. [1]

Uncertainty in the mean = ... N

(iii) The teacher states if the percentage uncertainty in the mean is **less than 10%** the data is repeatable.

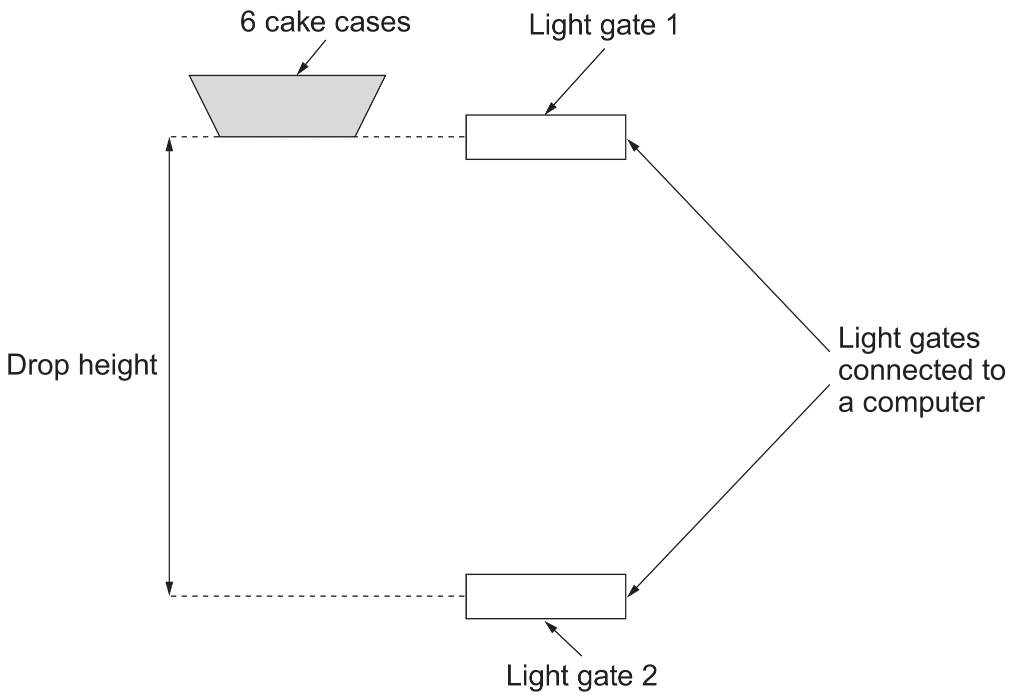
Use the equation:

$$\text{percentage uncertainty} = \frac{\text{uncertainty in the mean}}{\text{mean weight}} \times 100\%$$

to determine if the data collected is repeatable. [2]

....

(b) Light gates were used to time the falling cake cases.
The vertical distance between the two light gates was adjusted so that different drop heights could be investigated.
The apparatus is shown below.



Their results are shown in **Table 2** below.

Table 2

Drop height (m)	Time taken for the stack of 6 cake cases to fall (s)			Mean speed (m/s)
	Trial 1	Trial 2	Mean	
0.00				0.00
0.20	0.365	0.366	0.366	0.55
0.60	0.698	0.697	0.698	0.86
0.90	0.916	0.920	0.918	0.98
1.20	1.133	1.131	1.132	1.06
1.60	1.455	1.454	1.455	1.10
2.00	1.816	1.820	1.818	1.10
2.50	2.270	2.274	2.272	1.10
3.00	2.729	2.725	2.727	1.10

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(i) Give a reason why light gates, rather than a stopwatch, are used to measure the time taken for the stack of cake cases to fall. [1]

....

....

(ii) A student correctly notices that the mean speed of the cake cases at a drop height of 1.60 m is **double** the mean speed at a drop height of 0.20 m.
The student states that the cake cases have **4 times** more kinetic energy after falling 1.60 m compared to 0.20 m.

Use an equation from page 2 to investigate whether the claim is correct.
The total mass of the 6 cake cases is 3.3×10^{-3} kg. [3]
Space for calculations.

....

....

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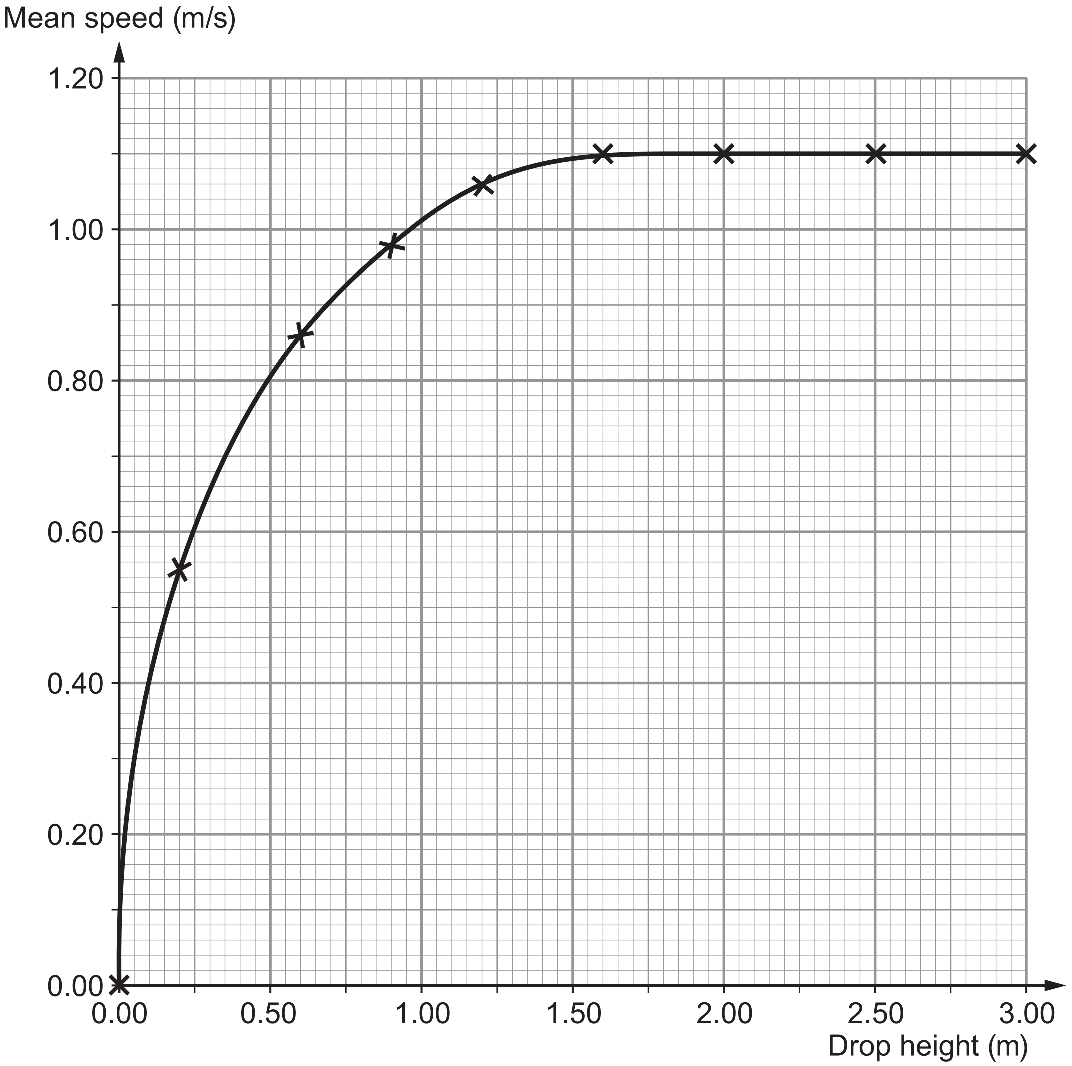
(c) A graph of mean speed against drop height for the 6 cake cases is shown below.



(i) Describe how the mean speed varies with drop height between **0.00 m and 1.50 m**. [2]

....

....

(ii) Use the graph to state the value of terminal speed for the cake cases. [1]

Terminal speed = ... m/s

(iii) I. Name the **two** forces acting on the falling cake cases. [1]

....

II. Compare the size of these two forces when the cake cases fall at terminal speed. [1]

....

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Turn over.